

Histidine-rich glycoprotein ameliorated lung ischemia-reperfusion injury in a mouse model¹



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KEYWORDS:

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BACKGROUND: Histidine-rich glycoprotein (HRG) is a multifunctional plasma glycoprotein involved in the regulation of various biological processes. Decreased plasma levels of HRG at 72 h after lung transplantation have been shown to be associated with severe primary graft dysfunction and poor transplant outcomes. In this study, we examined the effect of HRG supplementation on lung ischemia-reperfusion injury (IRI) in mice.

METHODS: The experimental mice were divided into a Sham group and three IRI groups (n=5 each) that received phosphate-buffered saline, human serum albumin, or HRG (HRG/IRI group), respectively. The lung function, lung injury, neutrophil infiltration, cellular apoptosis, inflammatory mediators, neutrophil extracellular traps (NETs), translocation status of high-mobility group box 1 (HMGB1), and hemolytic products in the injured tissue/plasma were compared among the groups.

RESULTS: The HRG/IRI group showed significantly improved oxygenation capacity and a reduced severity of lung injury as compared with the other IRI groups. The HRG/IRI group also showed

Abbreviations: DAMPs, damage-associated molecular patterns; HMGB1, high-mobility group box 1; HRG, histidine-rich glycoprotein; HSA, human serum albumin; IL, interleukin; IRI, ischemia-reperfusion injury; LT, lung transplantation; Ly6G, lymphocyte antigen 6 family member G; MPO, myeloperoxidase; NET, neutrophil extracellular trap; PBS, phosphate-buffered saline; PGD, primary graft dysfunction; TUNEL, terminal deoxynucleotidyl transferase-mediated deoxyuridine triphosphate nick-end labeling

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significantly less severe tissue inflammation in the reperfused lung, including a lower level of neutrophil infiltration, apoptotic cells, and neutrophil-associated inflammatory mediators than the other IRI groups. Moreover, formation of NETs and HMGB1 translocation were significantly suppressed in the HRG/IRI group as compared with the other IRI groups. A significantly greater degree of scavenging of hemolytic products was observed in the HRG/IRI group than the other IRI groups.

CONCLUSION: HRG supplementation ameliorated lung IRI in mice by suppressing inflammatory responses triggered by neutrophil-associated factors, including formation of NETs, HMGB1 translocation, and accumulation of hemolytic products.

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Introduction

Lung transplantation (LT) is offered as a last resort for patients with end-stage lung diseases. However, lung ischemia-reperfusion injury (IRI), a major cause of primary graft dysfunction (PGD) after LT, is associated with chronic lung allograft rejection and reduced survival after LT.¹ Lung IRI is associated with cellular stress, oxidative damage and inflammation, resulting in graft dysfunction and poor transplant outcomes.² Lung IRI involves multifaceted mechanisms, including formation of neutrophil extracellular traps (NETs)³ and release of damage-associated molecular patterns (DAMPs) such as high-mobility group box 1 (HMGB1)⁴ and hemolytic products.⁵ Among these, HMGB1-mediated inflammatory responses have been shown to be inhibited by histidine-rich glycoprotein (HRG).⁶

HRG, which was first isolated from the human serum and characterized in 1972,⁷ is a 75-kDa multifunctional plasma glycoprotein that is mainly produced in the liver and occurs relatively abundantly in the circulation.⁸ HRG is also expressed on the surfaces of leukocytes⁹ and maintains circulating neutrophils in a resting state to prevent vascular endothelial cells from uncontrolled activation.¹⁰ In fact, plasma HRG levels are decreased in mice with sepsis, whose survival has been shown to improve with HRG supplementation.¹⁰ Patients with sepsis showing decreased plasma HRG levels have also been demonstrated to show poor survival.¹¹

Consistent with the observations in patients with sepsis, we showed in a previous study that LT patients showing decreased plasma HRG levels at 72 h after LT developed severe PGD and showed decreased overall survival after LT.¹² Therefore, we hypothesized that HRG could offer promise as a potential therapeutic target for lung IRI. In this study, we examined the effect of HRG supplementation on lung IRI using a mouse hilar clamp model.

Materials and methods

Animals

Pathogen-free male C57BL/6J mice (CLEA Japan, Tokyo, Japan) weighing 20–25 g at 8–12 weeks were utilized for the experiments. All mice were cared for in accordance

with the principles of laboratory animal care formulated by the National Society for Medical Research and the Guide for the Care and Use of Laboratory Animals: Eighth Edition (National Academy Press, 2011). The research protocol was approved by the relevant ethics committee and institutional review board (OKU-2021661).

Purification of HRG from human plasma

HRG was purified from human plasma (supplied by the Japanese Red Cross Society), as described previously.¹³ In brief, human plasma was incubated with nickel-nitrilotriacetic acid agarose (Qiagen, Hilden, Germany) for 2 h at 4 °C under gentle shaking. The gel was packed in a column and washed sequentially with 10 mM Tris-buffered saline containing 10 mM imidazole (pH 8.0) and 10 mM Tris buffer containing 1 M NaCl (pH 8.0). Human HRG was eluted with 0.5 M imidazole in 10 mM Tris-buffered saline (pH 8.0). The protein extract was further purified on a Mono Q column (GE Healthcare, Little Chalfont, UK) with an NaCl gradient. Purified human HRG was identified by sodium dodecyl sulfate–polyacrylamide gel electrophoresis and Western blotting using human HRG-specific antibodies.

Lung IRI model

A mouse hilar clamp model was used as the lung IRI model (Figure 1), as previously described⁴ (see [Supplemental Material](#) for details). Animals in the Sham group (n = 5) underwent left thoracotomy without a hilar clamp and the healthy control group (n = 2) consisted of clean animals that were immediately sacrificed. Mice in the IRI groups were subjected to left lung ischemia for 1.5 h using a left hilar clamp, followed by 3 h of reperfusion. At sacrifice, arterial blood gases were measured, with oxygen delivered at a FiO₂ of 1.0 after clamping the right pulmonary hilum for 3 min.¹⁴

HRG supplementation

The mice scheduled to be subjected to IRI were administered 8 µl/g of vehicle (phosphate-buffered saline, PBS) (PBS/IRI group, n = 5), 20 mg/kg (8 µl/g) of protein

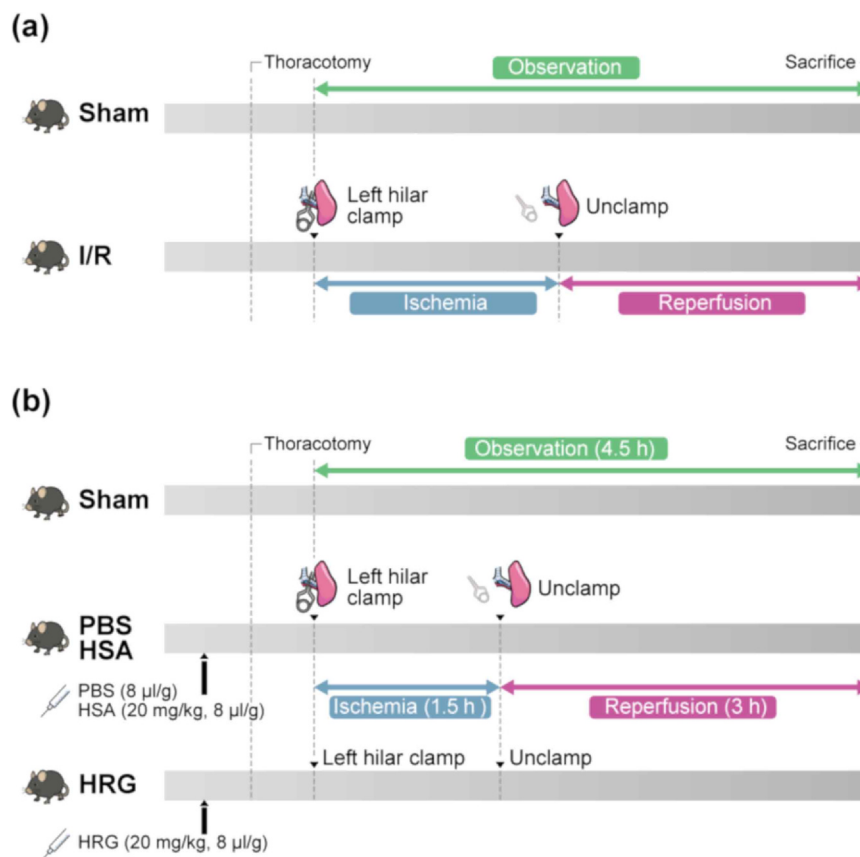


Figure 1 Experimental protocol for creation of a mouse model of lung IRI. (a) Schematic diagram of preliminary experiments. (b) Schematic diagram of treatment experiments.

control, namely, human serum albumin (HSA) (HSA/IRI group, $n = 5$), or 20 mg/kg (8 µl/g) of HRG (HRG/IRI group, $n = 5$) into the tail vein prior to the thoracotomy. The concentrations (8 µl/g) of HSA and HRG were adjusted to match the PBS dose per body weight, in accordance with the study protocol described previously.¹⁰

Histological lung injury score analysis

Histological lung injury was evaluated in a blinded manner by two pathologists (T. O. and A. M.), who assigned scores to the specimens based on the lung injury criteria: (1) neutrophil infiltration or aggregation; (2) airway epithelial cell damage; (3) interstitial edema; (4) hyaline membrane formation; and (5) hemorrhage.¹⁵ The severity of each pathological feature was assigned a score of 0, 1, 2, 3 or 4, corresponding to absent, minimal, mild, moderate, or severe injury, respectively. The total score obtained by adding the scores for the individual pathological features represented the total lung injury score.

Analysis of the severity of apoptosis

The apoptosis severity was analyzed by terminal deoxynucleotidyl transferase-mediated deoxyuridine triphosphate nick-end labeling (TUNEL), in accordance with the

manufacturer's instructions (see [Supplemental Material](#) for details).

Quantification of cytokines and chemokines

The concentrations of cytokines and chemokines in the plasma and lung samples were measured using Bio-Plex Pro Mouse Cytokine 23-plex Panel (Bio-Rad, CA, USA), in accordance with the manufacturer's instructions.

Immunohistochemistry

Immunohistochemical analysis was performed using anti-HMGB1 pAb (#ab18256, Abcam, Cambridge, UK), anti-myeloperoxidase (MPO) pAb (#AF3667, R&D systems, MN, USA), anti-histone H3 pAb (#ab5103, Abcam), and anti-lymphocyte antigen 6 complex locus G6D (Ly6G) mAb (#127602, Biolegends, CA, USA), as described previously^{4,10} (see [Supplemental Material](#) for details).

NETs are defined as extracellular structures showing colocalization of MPO and histone H3, and are detected by immunofluorescence within 100 µm beneath the pleura. NET-positive sites were counted in three high-power fields and their average values were determined, as described previously.¹⁰ Cells with positive HMGB1 expression outside the nuclei were counted as cells showing HMGB1 translocation from the nuclei, and the average percentages

of these cells relative to the total number of cells in three high-power fields were calculated.

Enzyme-linked immunosorbent assay

The protein levels of HMGB1 in the plasma and lung samples were evaluated using the HMGB1 ELISA Kit Exp (Shino-Test Co., Sagamihara, Japan), in accordance with the manufacturer's instructions.

Western blotting

Cryopreserved lung tissues were subjected to Western blotting, as previously described⁴ (see [Supplementary Material](#) for details), using rabbit anti-HRG pAb (#GTX64492, Gene Tex, CA, USA), rabbit anti-haptoglobin (#ab1312369, Abcam), rabbit anti-hemopexin mAb (#ab150401, Abcam), and goat anti-hemoglobin (#GTX77484, Gene Tex).

Statistical analysis

Prism 9 (GraphPad Software, CA, USA) was used for the statistical analyses. Statistical comparisons between two groups were performed using Student's t-test and comparisons among multiple groups were performed by one-way ANOVA followed by the post hoc Šídák's test for multiple comparisons. $P < 0.05$ was considered as denoting significance.

Results

Decrease of plasma HRG levels in the murine lung IRI

First, we performed preliminary experiments to determine the optimal duration of ischemia and reperfusion for obtaining appropriate mouse models of lung IRI from among different time combinations ([Figure 1a](#)). The combination of 1.5-hour ischemia and 3-hour reperfusion was identified as the optimal protocol based on significant differences observed in the following features between the groups ([Figure 1b](#)). Diffuse lung redness on gross inspection was observed in the IRI group as compared with the Sham group ([Figure 2a](#)). The lung injury scores were significantly higher in the IRI group than in the Sham group ($P = 0.0085$; [Figure 2b, c](#)), especially the score for hemorrhage ($P = 0.0002$). The oxygenation capacity was significantly worse in the IRI group than in the Sham group ($P = 0.0005$; [Figure 2d](#)). Notably, the plasma HRG levels were significantly lower in the IRI group than in the Sham group ($P = 0.0011$; [Figure 2e](#)).

HRG supplementation ameliorated the lung IRI

[Figure 3](#) shows the results of the treatment experiment. The HRG/IRI group showed a pinkish white appearance of the reperused lungs as compared with the PBS/IRI and HSA/IRI groups, similar to the Sham group ([Figure 3a](#)). The HRG/IRI group showed significantly lower lung injury

scores, especially the scores for neutrophil infiltration and hemorrhage, than the PBS/IRI ($P = 0.0028$) and HSA/IRI ($P = 0.0003$) groups ([Figure 3b, c](#)). The HRG/IRI group also showed significantly better oxygenation capacity than the PBS/IRI and HSA/IRI groups ($P < 0.0001$ for both) ([Figure 3d](#)). The results of TUNEL staining performed to evaluate the severity of apoptosis as a component of tissue damage ([Figure 3e](#)) revealed significantly fewer TUNEL-positive cells in HRG/IRI group than in the PBS/IRI ($P = 0.0011$) and HSA/IRI ($P = 0.0002$) groups. Cytokine assay of the lung tissue revealed significantly lower levels of neutrophil-associated chemokines ([Figure 3f](#)), including C-X-C motif chemokine ligand 1, granulocyte colony-stimulating factor, and granulocyte-macrophage colony-stimulating factor, in the HRG/IRI group than in the PBS/IRI ($P = 0.0033$, 0.0464 , and 0.0012 , respectively) and HSA/IRI ($P = 0.0200$, 0.0482 , and 0.0439 , respectively) groups. Significant differences were also observed in the levels of macrophage inflammatory protein-1 α ($P = 0.0373$), 1 β ($P = 0.0297$), interleukin (IL)-6 ($P = 0.0173$), and IL-10 ($P = 0.0043$) between the HRG/IRI and PBS/IRI groups. In contrast, there was a significant difference in the level of IL-1 β only between the HRG/IRI and HSA/IRI groups ($P = 0.0247$). Furthermore, no significant differences in the tissue levels of tumor necrosis factor- α were observed among the groups.

HRG suppressed neutrophil-associated inflammatory responses

The results of immunohistochemical analysis revealed that, similar to the Sham group, the HRG/IRI group showed a significantly lower degree of neutrophil infiltration, which was identified as the sum of MPO- and Ly6G- positive cells, than the PBS/IRI and HSA/IRI groups (MPO, $P < 0.0001$ for both; Ly6G, $P = 0.0014$ and $P < 0.0001$, respectively) ([Figure 4a, b](#)). A correlation diagram between MPO- and Ly6G- positive cells is shown in [Figure 4c](#). Neutrophil-derived MPO activity was suppressed in the HRG/IRI group as compared with the PBS/IRI and HSA/IRI groups.

Evaluation of NETs by fluorescence immunostaining ([Fig. 4d](#)) revealed that the overall number of NETs was small and that the NETs were mainly located just beneath the pleura. A significantly smaller number of NETs per unit area within 100 μm beneath the pleura was observed in the HRG/IRI group than in the PBS/IRI ($P = 0.0082$) and HSA/IRI ($P = 0.0230$) groups.

HRG supplementation decreased HMGB1 translocation

Immunofluorescence staining showed that HMGB1 was localized in the nuclei in the Sham group, whereas nuclear HMGB1 was translocated to the cytoplasm and then to the extracellular spaces in the PBS/IRI and HSA/IRI groups ([Figure 5a](#)). This translocation of HMGB1 was significantly suppressed in the HRG/IRI group as compared with that in the PBS/IRI and HSA/IRI groups ($P < 0.0001$ for both).

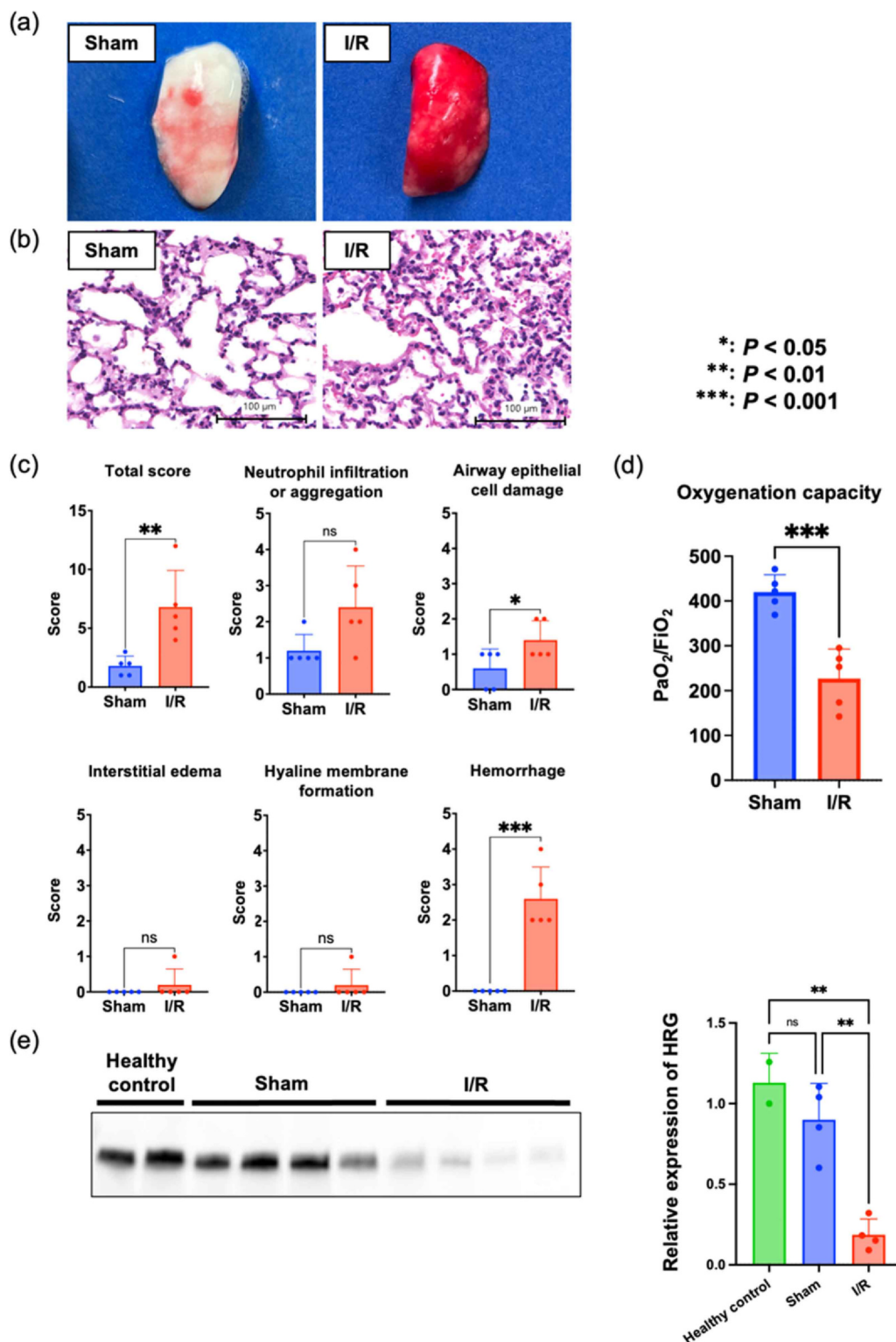


Figure 2 Results of preliminary experiments performed with an ischemia time of 1.5 h and reperfusion time of 3 h. (a) Representative images of gross appearance of the left lung. (b) Representative images of hematoxylin and eosin staining. Scale bars represent 100 μ m. (c) Total lung injury score and scores for each of the injury criteria. (d) Lung oxygenation capacity as assessed by the PaO₂/FiO₂ ratio. (e) HRG levels in plasma samples analyzed by Western blotting. Data are expressed as means $\pm$ standard error of the mean (SEM), * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

HRG decreased the accumulation of hemolysis-derived DAMPs

Plasma markers of hemolysis were evaluated by Western blotting (Fig. 5b). As a result of adequate HRG

supplementation, the plasma HRG levels were relatively high in the HRG/IRI group and significantly lower in the PBS/IRI and HSA/IRI groups ($P < 0.0001$ for both), consistent with the results of the preliminary experiments. The plasma levels of haptoglobin and hemopexin, which are scavengers of hemolytic

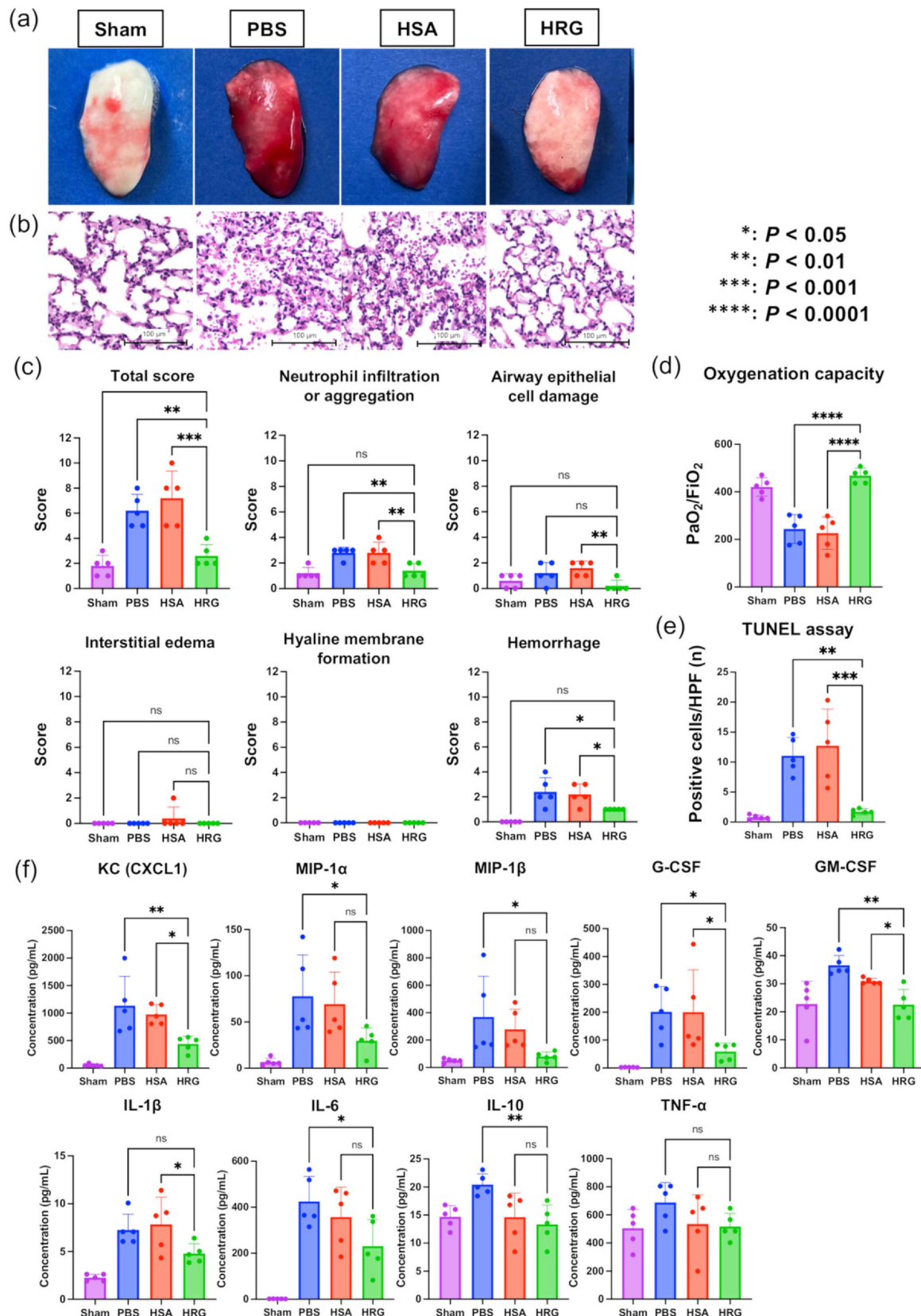


Figure 3 Results of treatment experiments performed with an ischemia time of 1.5 h and reperfusion time of 3 h. (a) Representative images of gross appearance of the left lungs in representative cases. (b) Hematoxylin and eosin staining in representative cases. Scale bars equal to 100 μ m. (c) Total lung injury score and scores for each of the criteria of IRI. (d) Lung oxygenation capacity as assessed by the PaO₂/FiO₂ ratio. (e) Number of TUNEL-positive apoptotic cells assessed by immunostaining. (f) Concentrations of cytokines and chemokines in the lung tissue by a multiplex immunoassay. Data are expressed as the means $\pm$ SEM, * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$.

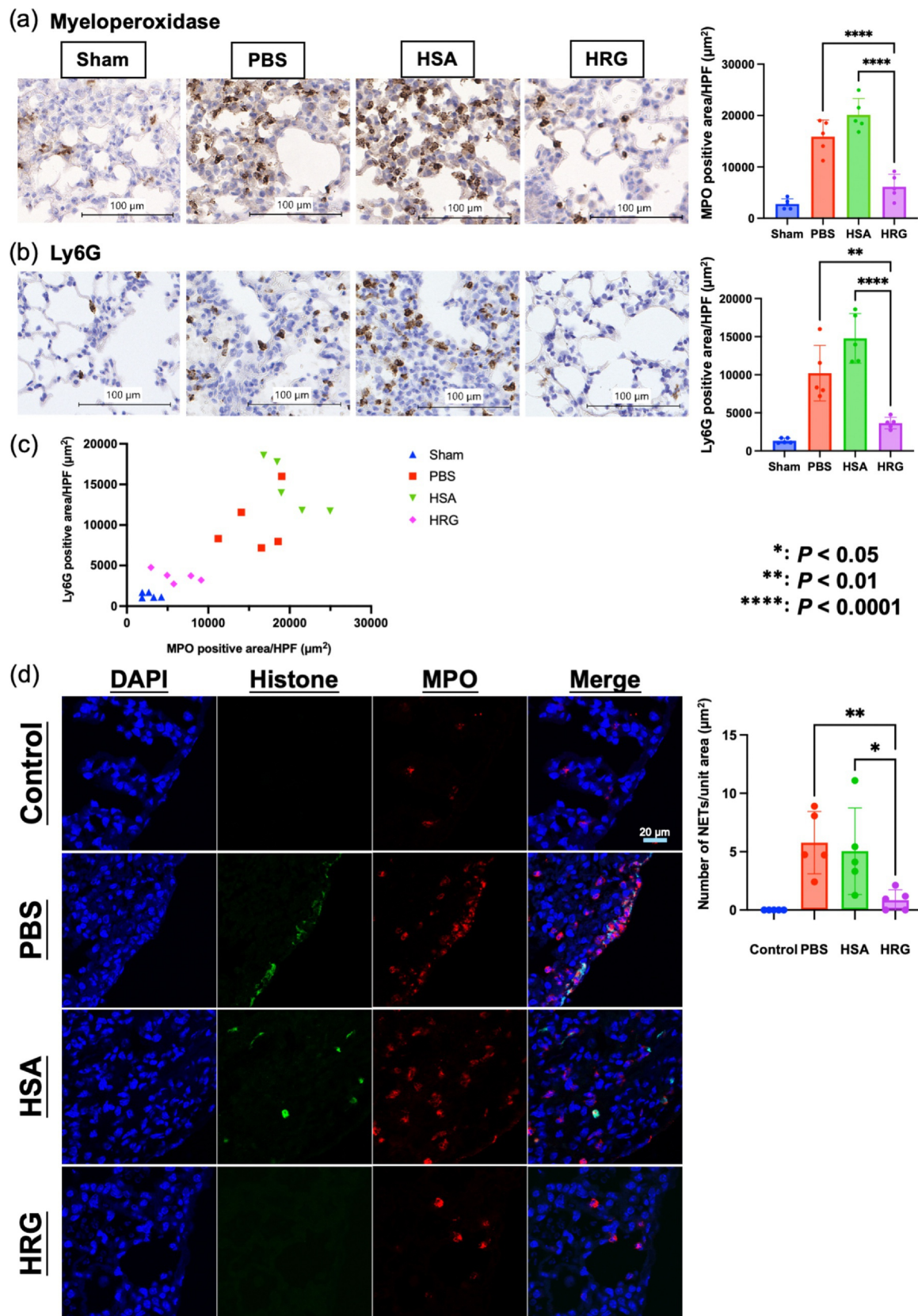


Figure 4 (a, b) Representative images of immunostaining for MPO and Ly6G with quantification of the area occupied by cells showing positive staining for MPO and/or Ly6G. Scale bars represent 100 μm . (c) A correlation diagram of the areas occupied by MPO- and Ly6G-positive cells. (d) Representative images of fluorescence immunostaining for NETs and evaluation of the number of NETs. Scale bars represent 20 μm . Data are expressed as the means $\pm$ SEM, * $P < 0.05$, ** $P < 0.01$, **** $P < 0.0001$.

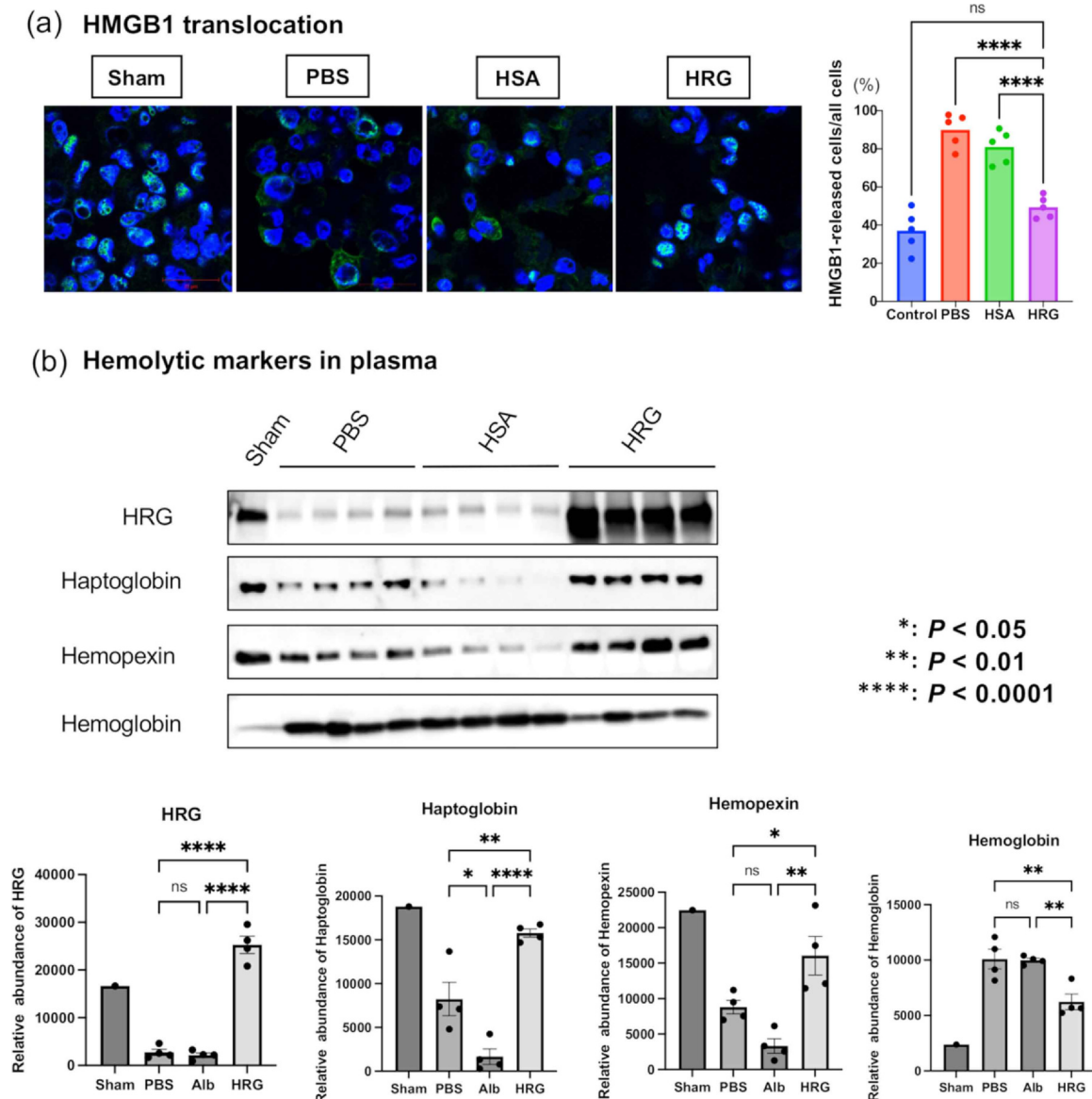


Figure 5 (a) Representative images of fluorescence immunostaining for HMGB1 and evaluation of the number of cells showing HMGB1 translocation from the nucleus. Scale bars represent 20 μ m. (b) Hemolytic marker levels in the plasma samples were analyzed by Western blotting. Data are expressed as the means $\pm$ SEM, * $P < 0.05$, ** $P < 0.01$, **** $P < 0.0001$.

products, were maintained in the HRG/IRI group, as in the Sham group, while they were significantly decreased in the PBS/IRI ($P = 0.0052$ and 0.0429 , respectively) and HSA/IRI groups ($P < 0.0001$ and 0.0017 , respectively) as compared with the HRG/IRI group. In contrast, the level of free hemoglobin, which is known as a hemolysis-derived DAMP, was significantly decreased in the HRG/IRI group as compared with the PBS/IRI ($P = 0.0065$) and HSA/IRI ($P = 0.0077$) groups.

Discussion

In this study, we found significantly reduced plasma HRG levels in murine lung IRI, consistent with the results of a previous clinical study¹². HRG supplementation alleviated lung IRI in mice, with decreases in the degree of neutrophil infiltration, severity of apoptosis, expressions of neutrophil-

associated inflammatory mediators, NET formation, HMGB1 translocation, and production of hemolysis-derived DAMPs. Our results suggest that HRG supplementation may offer promise as a potential therapeutic strategy for lung IRI.

Plasma HRG levels decrease during severe systemic inflammation, and HRG supplementation has been shown to improve outcomes in septic mice.¹⁰ Similarly, in a mouse model of liver IRI with non-alcoholic steatohepatitis, HRG-knockdown mice showed severe liver damage, which was effectively prevented by HRG supplementation.¹⁶ Furthermore, congenital HRG deficiency is a risk factor for cerebral infarction and thrombophilia.¹⁷ Given these previous findings, we considered that lung IRI, associated with decreased plasma levels of HRG, may be potentially treatable by HRG supplementation. To the best of our knowledge, this is the first report focusing on the role of HRG in lung IRI.

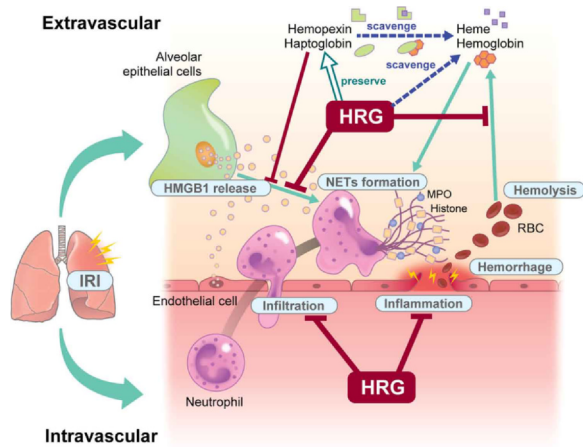


Figure 6 Summary of the multifaceted mechanisms underlying the efficacy of HRG against lung IRI inferred from this study.

Our results showed that HRG supplementation suppressed the degree of neutrophil infiltration, release of neutrophil-associated cytokines, and formation of NETs in the murine lung IRI. HRG has been shown to control inflammation by maintaining the neutrophil morphology and inhibiting both neutrophil adhesion to endothelial cells and formation of NETs.^{10,16} Neutrophil infiltration and the release of neutrophil-associated cytokines are the main features of lung IRI,² and formation of NETs has also been shown to be increased in the lung tissue in murine lung IRI.³ Increased numbers of NETs have been shown in the alveolar lavage fluid of patients with PGD after LT, and to be associated with the development of PGD,³ suggesting that NETs could be possible therapeutic targets for PGD. Taken together, suppression of the formation of NETs would be one of the key mechanisms by which HRG supplementation ameliorated lung IRI in mice.

Among the DAMPs, extracellular HMGB1 directly contributes to the production of inflammatory cytokines and development of IRI.¹⁸ Hemolytic products, including free hemoglobin, heme, and divalent iron ions, also represent DAMPs.^{19,20} Under sepsis conditions, HRG effectively inhibits the HMGB1 translocation from the nucleus to the cytoplasm and extracellular space, contributing to the suppression of hemolysis by protecting the vascular endothelial barrier function.⁶ Translocation of nuclear HMGB1 to the cytoplasm and extracellular space, which was previously reported in a brain IRI model,²⁰ was observed in the lung tissue of murine lung IRI in this study, consistent with previously reported results.⁴ Notably, HMGB1 translocation was significantly suppressed in the HRG/IRI group in this study. Moreover, the level of free hemoglobin was significantly decreased, and the levels of haptoglobin and hemopexin, which are known as scavengers of hemolytic products, were significantly increased in the HRG/IRI group as compared with the other groups. HRG promotes scavenging of hemolytic products such as DAMPs by avidly binding to them, effectively preventing sequential hemolysis.²¹ To prevent their detrimental effects, hemolytic products are scavenged by acute-phase plasma proteins,

e.g., extracellular hemoglobin by haptoglobin and extracellular heme by hemopexin, and rapidly removed from the circulation.¹⁹ Considering these observations, our results suggest that HRG may act as a scavenger of hemolytic products while sparing haptoglobin and hemopexin in the plasma.

Based on the results of previous studies and this study, we speculate the involvement of the following multifaceted mechanisms to explain the therapeutic efficacy of HRG in lung IRI (Figure 6). HRG inhibits neutrophil infiltration from vascular endothelial cells and suppresses the release of HMGB1 from the nuclei of the vascular endothelial and alveolar epithelial cells, as well as prevents the formation of NETs associated with hemorrhage and hemolysis. Simultaneously, HRG decreases vascular inflammation, contributing to reduction of hemorrhage and hemolysis. In addition, in combination with hemopexin and haptoglobin, HRG acts as a scavenger for hemolysis-derived DAMPs. Therefore, depletion of HRG might impair biological regulation and exacerbate lung IRI through these multifaceted mechanisms.

In a previous clinical study, patients with severe PGD were found to have significantly lower plasma HRG levels than those with no or moderate PGD at 72 h after LT, and high plasma HRG levels at 72 h after LT were associated with improved overall survival.¹² Given the results of this study, HRG supplementation might alleviate lung IRI and severe PGD, presumably contributing to improved survival after LT. Regarding clinical application of HRG, although HRG has the advantage of an established purification protocol among numerous types of plasma proteins,¹³ the amount of available HRG is limited due to its dependence on blood donation. To resolve this problem, further study is required to develop recombinant HRG protein or to identify a low-molecular-weight protein to which HRG can bind.

This study had several limitations. To minimize experimental errors, we used a hilar clamp model and not a LT model, because of its simplicity and high reproducibility. Since the lung graft is exposed to warm and not cold ischemia in this model, this model does not reproduce the clinical condition of IRI after LT from donors after brain death, in contrast to donors after circulatory death. Although HRG was administered before the thoracotomy in this study, administration after reperfusion may be more therapeutic and more clinically relevant for IRI after LT. Further, the optimal route of administration could differ in LT, including intravenous administration, administration by mixing in the perfusion solution injected into the graft pulmonary artery or preservation solution for the graft, or ex-vivo lung perfusion. Our study did not include assessments using transgenic or knockout mice, and the precise mechanism of HRG action should be elucidated in future studies employing genetically modified animals in a mouse LT model. Further research using rodent or porcine LT models is needed to assess the clinical feasibility of HRG supplementation for lung IRI after LT. In addition, because this study relied on a single experimental model with a limited sample size, and our histological evaluation tended to focus on hemorrhagic findings, along with the absence of

vascular permeability assessment, the generalizability and interpretability of our results may remain constrained.

In conclusion, HRG supplementation ameliorated lung IRI in mice, by suppressing the inflammatory responses caused by neutrophil-associated factors, including formation of NETs, HMGB1 translocation, and accumulation of hemolysis-derived DAMPs. Although the mechanism of HRG action needs to be elucidated in other LT models, HRG supplementation might offer a potential therapeutic strategy for lung IRI.

CRediT authorship contribution statement

Y.K., S.S., and D.O. designed and performed the experiments, analyzed and interpreted the data. Y.K. and S.S. drafted and revised the manuscript. D.W. performed the experiments and analyzed the data. T.O. and A.M. performed histological analyses. All authors have read and approved the final version of the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplemental data associated with this article can be found in the online version at [doi:10.1016/j.jhlto.2025.100453](https://doi.org/10.1016/j.jhlto.2025.100453).

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